

Exocrine Pancreatic Insufficiency After Allogeneic Bone Marrow Transplantation

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Summary

Several reports suggest a relationship between bone marrow transplantation and pancreatic damage. The authors describe two patients with pancreatic insufficiency after bone marrow transplantation. The first patient had weight loss arising from steatorrhea secondary to severe pancreatic insufficiency (lipase output result <3% of normal) due to pancreatic atrophy. The second patient had steatorrhea secondary to intestinal bacterial overgrowth, but moderate pancreatic insufficiency was present (results of trypsin and lipase outputs were 50 and 30% of normal). Before the diagnosis of pancreatic insufficiency, both patients had acute graft vs host disease and the second patient had chronic graft vs host disease. We conclude that graft vs host disease after bone marrow transplantation may damage the pancreas. Patients with persistent malabsorption after bone marrow transplantation should be tested for exocrine pancreatic insufficiency.

Key Words: Marrow transplantation; graft vs host disease; exocrine pancreatic insufficiency.

Introduction

Most often chronic pancreatic insufficiency is secondary to primary disease of the pancreas. The most common cause of primary pancreatic disease is chronic pancreatitis, but pancreatic cancer and cystic fibrosis are also relatively common causes (1). The most common secondary causes of pancreatic insufficiency are small intestinal mucosal disease (nontropical sprue) (2) and Zollinger Ellison Syndrome (1). Drugs probably do not cause pancreatic insufficiency (1,3), but there are several reports of an association between pancreatic disease and bone marrow

transplantation (BMT) (4–7) as well as after radiation therapy (8–10).

The aim of this report is to describe two patients who developed exocrine pancreatic insufficiency after allogeneic bone marrow transplantation.

Case 1

A 22-yr-old white woman was admitted to our institution who had an 8-mo history of five to six watery stools daily that eventually became greasy. She also had nausea and weight loss of 8 kg.

Fourteen months before the onset of diarrhea the patient was admitted to another hospital where she had a diagnosis of acute myeloid leukemia (M2). After induction therapy with daunorubicin and cytarabine complicated by non-A, non-B hepatitis, she relapsed. Subsequently, she had an allogeneic BMT (HLA-matched brother). When she presented to us

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she was taking prednisone for acute graft vs host disease (GVHD) of the liver and skin. There were no signs of chronic GVHD. She had no history of alcohol abuse.

On physical examination she appeared chronically ill. There were no masses or organomegaly. Pertinent laboratory data included a 48-hr stool collection of 83 g of fat per 24 h (normal result $<7 \text{ g}^{-24 \text{ h}}$); 58% of her stool was fat (normal result $<20\%$). Stools had no leukocytes or parasites and results of the stool cultures were normal. A duodenal aspirate yielded no bacterial overgrowth or giardiasis. Her serum carotene was low ($17 \text{ mg}^{-\text{dL}}$; normal $48\text{--}200 \text{ mg}^{-\text{dL}}$). The result of the cholecystokinin octapeptide pancreatic function test (11–13) was markedly abnormal; the trypsin output was $1 \cdot 10^3 \text{ U}^{-\text{h}}$ (normal $25\text{--}54 \cdot 10^3 \text{ U}^{-\text{h}}$) and lipase output was $1 \cdot 10^3 \text{ U}^{-\text{h}}$ (normal $77\text{--}322 \cdot 10^3 \text{ U}^{-\text{h}}$). The results of the following tests were normal: hemoglobin, white blood cell count, platelet count, prothrombin time, urinalysis, and serum concentrations of B12, folate, sodium, potassium, calcium, phosphorus, total protein, glucose, alkaline phosphatase, AST, bilirubin, uric acid, creatinine, thyroxine, gastrin, and VIP. Computed tomographic (CT) scans of her abdomen showed only a trivial amount of pancreatic tissue in the head of the pancreas. The clinical diagnosis was chronic pancreatic insufficiency secondary to pancreatic atrophy. Therapy was a low fat diet and oral pancreatin. After 4 yr the patient is alive and well, working full-time, and continues to take Pancreatin. There is no GVHD and the results of the hematology tests are normal.

Case 2

A 38-yr-old white woman was admitted to our Institution with massive splenomegaly and Philadelphia Chromosome positive chronic granulocytic leukemia. She had no history of alcohol abuse.

After reduction of leukocytes and splenomegaly with hydroxyurea she had an allogeneic BMT (HLA-matched sister). She experienced minimal acute GVHD consisting of grade I skin GVHD that cleared after 2 wk of prednisone therapy. Prophylaxis for chronic GVHD consisted of standard short course methotrexate and cyclosporine. One hundred seventy days after the transplant, she developed

severe chronic GVHD of the skin and liver complicated by histologically proved drug-induced hepatitis, presumed to be secondary to azathioprine. Azathioprine was discontinued, cyclosporine was resumed, and prednisone was continued.

For the next 3 mo she gradually lost weight (8.0 kg). She was readmitted and a stool exam was consistent with steatorrhea. Results of gastroscopy were consistent with chronic GVHD and included esophageal candidiasis and superficial erosions of the fundus and body of the stomach. Colonoscopy was normal. The serum carotene ($18 \text{ ng}^{-\text{dL}}$), copper ($0.56 \text{ } \mu\text{g}^{-\text{mL}}$), magnesium ($1.59 \text{ mg}^{-\text{dL}}$), and zinc ($0.27 \text{ } \mu\text{g}^{-\text{mL}}$) were low and the prothrombin time was prolonged (13.3 s). Culture of the duodenal aspirate revealed bacterial overgrowth with streptococci and gram-negative cocci. The findings of a small bowel biopsy were normal. Pancreatic enzyme secretion in response to the cholecystokinin octapeptide pancreatic function test was low; trypsin and lipase were 14 and $24 \cdot 10^3 \text{ U}^{-\text{h}}$, respectively. The pancreas was interpreted as being normal on abdominal ultrasound.

An endoscopic percutaneous gastrostomy was performed and enteral nutrition was initiated. The clinical diagnosis was chronic GVHD with cutaneous, liver, esophageal involvement; steatorrhea secondary to small bowel bacterial overgrowth, idiopathic pancreatic insufficiency. After 3 mo of enteral feeding, prednisone, and cyclosporine, the patient gained 5 kg. At this point the patient resumed oral alimentation and the gastrostomy tube was removed. After 6 mo, liver enzymes increased to abnormal levels and severe diarrhea recurred, which did not abate by taking tetracycline or pancreatic enzyme supplements separately for a month each. Eventually, the patient gained weight to 56 kg, but still has one to six stools a day with a daily caloric intake of 4500–6000 cal.

Discussion

BMT is a treatment for severe aplastic anemia, leukemia, lymphoma, and various congenital disorders affecting the hematologic system. With increasing numbers of transplants, improving chemotherapy and radiation treatments, and methods to prevent and manage posttransplant complications, the population of long-term survivors is growing steadily and newly

associated diseases are being discovered in these patients. We report two women with leukemia who had a BMT 6 and 8 mo before admission to our institution for evaluation of weight loss and steatorrhea.

In a recent large retrospective study of long-term survivors of allogeneic BMT (14), weight loss ranked as the most frequent nutritional problem (present in 54 of 192 patients) but steatorrhea was diagnosed in only nine patients. In this study, steatorrhea was mainly secondary to intestinal bacterial overgrowth, cholestatic liver disease, or diffuse intestinal involvement secondary to GVHD. These authors did not determine the exact pathogenesis in individual patients, so pancreatic insufficiency was not excluded in some of these patients.

Another nutritional problem is malabsorption of vitamin B12. Absorption of this vitamin is severely reduced after alloageneic transplantation and is likely because of the effects of conditioning and GVHD on the small intestinal mucosa. The latter produces long-term impairment of vitamin B12 absorption (15).

Of note, patients receiving total body radiation and high dose chemotherapy (conditioning) for bone marrow transplant commonly have nausea, abdominal pain, and vomiting. These problems usually are attributed to this therapy and the associated mucositis. However, these symptoms are consistent with pancreatitis and testing is recommended in all. Pancreatic isoamylase elevations occur in 27% of patients following regimens that include total body radiation. These elevations do not return to preirradiation levels for a median of 1.5 mo (16).

In an autopsy series of 11 patients with engrafted allogeneic bone marrow transplantation, Nakamura (17) found lesions that could be associated with pancreatic insufficiency. Six of the 11 patients had intralobular fibrosis; of them, two had lymphocytic infiltration and one had atrophy of acini. An additional patient without fibrosis had focal necrosis of the pancreas. These lesions were in patients with acute GVHD. The lymphocytic infiltration was similar to experimental models of acute GVHD (18).

Conditioning may be related to posttransplant pancreatitis particularly in children (19). Werlin et al. (4) reported seven children (3.5%) with symptoms or laboratory evidence of acute pancreatitis, mainly in the first weeks after bone marrow trans-

plantation. All patients were receiving corticosteroids and Cyclosporin and five of seven patients also had GVHD. Pancreatitis also has been reported in patients receiving a mismatched marrow transplant (20), likely because GVHD occurs with a high incidence in this setting. Jurges et al. (5) reported a 7-yr-old boy with diarrhea and severe reduction of trypsin and lipase output 48 d after transplant. An edematous pancreas was seen on abdominal ultrasonography. Pancreatic enzyme supplements were started and steroids administered for skin, gastrointestinal, and hepatic GVHD. Seven months later symptoms disappeared and pancreatic supplements were discontinued. Tyden et al. (6) found mild elevation of pancreatic isoamylase in the first weeks after transplantation in 20 patients with nonspecific symptoms, but they did not investigate pancreatic function. Finally, Anderson (7) described a 40-yr-old woman with pancreatic insufficiency diagnosed by a pancreolauryl test. All these patients had acute GVHD and the last patient (7) also had chronic GVHD.

Our first patient had weight loss secondary to steatorrhea owing to a severe pancreatic insufficiency (lipase result <3% of normal); pancreatic steatorrhea occurs when lipase secretion is less than 10% of normal (11). Other causes of steatorrhea were excluded in this patient. She experienced acute, but not chronic GVHD.

The second patient had steatorrhea secondary to intestinal bacterial overgrowth. Moderate pancreatic insufficiency was present, but was not likely involved in the pathogenesis of malabsorption. Weight loss in this patient was due to chronic GVHD of esophagus and liver and malabsorption secondary to intestinal bacterial overgrowth. She had only minimal acute GVHD (grade I skin GVHD).

Pancreatic insufficiency may be present in patients undergoing allogeneic BMT because pancreatic disease was present before BMT, may be directly associated with the primary disease, may be related to BMT, or be coincidental. The first hypothesis is unlikely because the two conditions are rare and their association is improbable. Furthermore, neither patient had abdominal pain, which is a usual clinical presentation of chronic pancreatitis. In our experience young persons with idiopathic pancreatitis always have pain (1,21) and only 10–20% of patients with alcoholic pancreatitis never have pain. Finally,

it is unlikely that an atrophic pancreas seen on CT scan and associated with severe depletion of enzyme secretion was present before BMT. Malabsorption is always a late event in the natural history of chronic pancreatitis and it may take 10–20 yr to develop (our patient was 22-yr-old). There is no evidence in the medical literature of a relationship between leukemia and pancreatic insufficiency (1).

BMT causes esophageal, liver, intestinal (22–24), and perhaps pancreatic damage by three mechanisms:

1. Toxicity from pretransplant chemoradiation therapy with high dosage cytotoxic and immunosuppressive drugs and supralethal total body irradiation;
2. A unique immunologic disorder known as GVHD; and
3. Infection developing in a severely immunosuppressed host.

Several cancer chemotherapy agents including doxorubicin and cytarabine (but not hydroxyurea) have been associated with acute pancreatitis, but neither of our patients had any clinical, laboratory, or imaging signs suggesting acute pancreatitis. There are four case reports in the literature relating pancreatic damage to radiation (8–10).

In allogeneic transplantation, the donor and recipient are not genetically identical. The donor is usually a sibling and has some but not all of the same parental antigens as identified by human leukocyte antigen typing as the recipient. GVHD occurs when immunologically competent donor marrow cells, presumably T-lymphocytes, recognize recipient antigenic differences, and cause damage to tissue. Acute GVHD affects the skin, liver, and/or intestinal tract and typically occurs 20–100 d after transplant. Chronic GVHD, on the other hand, is a multisystem disorder that may follow acute GVHD or present *de novo*. Signs and symptoms may appear 70–400 d after transplant. Chronic GVHD is a major medical problem beyond 100 d after transplant, with an estimated incidence of 35–40%. The target organs of chronic GVHD are the skin, liver, mouth, esophagus, eye, vagina, and occasionally, the pulmonary, intestinal, and musculoskeletal system (14,22–24).

Both our patients had signs of acute GVHD and the second patient also had chronic GVHD; our data do not definitely suggest that GVHD has a role in the

pathogenesis of pancreatic insufficiency. However, this is the most plausible hypothesis.

Patients after BMT experience a high rate of infection attributable to persistent humoral and cell-mediated immune deficiencies as well as immunosuppressive therapy. Pulmonary, sinus, and systemic bacterial infections are the most frequent (14,22–24). Mumps, infectious hepatitis, and Coxsackie B viruses cause acute pancreatitis, but it is controversial if other infectious agents do (1). There is no evidence in the literature that an infectious agent causes chronic pancreatitis, but we cannot exclude that a subclinical infectious disease in immunocompromised patients may cause occult pancreatic damage.

In conclusion, the authors report pancreatic insufficiency in two patients who had a BMT. We postulate that the BMT may have a role in the pathogenesis of the pancreatic damage, probably secondary to GVHD. We suggest that patients with persistent malabsorption after BMT should be tested to exclude exocrine pancreatic insufficiency.

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